

VoteChain Project – Phase 2 Accelerated Recovery Plan

Duration: 13 November – 1 December 2025 Time Allocation: 1 hour/day per team member

Goal: Complete Phase 0 + Phase 1 + Phase 2, deliver working registration + voting prototype for 2nd installment submission.

Team Task Split & Daily Goals

Core Devs (Achieve + Rajib) - 13–15 Nov: Learn blockchain basics (Block, Transaction, Blockchain class) + pseudocode - 16–18 Nov: Implement Registration Module: Permanent hash (SHA256) + database storage - 19–21 Nov: Implement Temp Hash: tempHash = SHA256(permanentHash + electionName + date + salt) - 22–24 Nov: Voting module logic: scan fingerprint → generate tempHash → check duplicates → store vote - 25–27 Nov: Dead Block handling & duplicate vote prevention - 28–30 Nov: Testing all modules together (registration + voting + hash verification)

Frontend (Abhash) - 13–16 Nov: Registration page: Name, DOB, Auto VoterID, Fingerprint upload, Photo - 17–20 Nov: Voting page: Scan fingerprint → show voter info → select candidate → submit - 21–24 Nov: Connect frontend forms to backend API endpoints - 25–30 Nov: Display vote confirmation + temporary dashboard for votes

Backend (Sandesh) - 13–16 Nov: Setup Node.js/Flask project + database schema (voter info, fingerprints) - 17–20 Nov: /registerVoter endpoint: save permanent hash + voter info - 21–24 Nov: /verifyFingerprint + /castVote endpoints: handle tempHash voting - 25–30 Nov: Test full backend integration with frontend + core dev module

Testing & Documentation (Ayush) - 13–16 Nov: Prepare test templates for registration & voting - 17–24 Nov: Run unit tests for permanent hash, temp hash, duplicate votes - 25–30 Nov: Compile screenshots + prepare Phase 2 submission report

Deliverables by 1 Dec 2025

1. Working Registration Module: permanent hash + fingerprint + voter info
2. Working Voting Module: temp hash + duplicate vote prevention + Dead Block handling
3. Frontend pages: registration + voting + vote confirmation
4. Backend APIs: /registerVoter, /verifyFingerprint, /castVote
5. Testing logs + screenshots
6. Phase 2 Report: research + methodology + tools + prototype description

Notes for Success - Focus on simulation first: if fingerprint SDK is delayed, use dummy IDs for tempHash tests. - Core Devs + Backend must coordinate daily: temp hash → DB → frontend. - Screenshots & logs daily for easy submission at the end. - Strictly follow 1 hr/day schedule to complete all phases on time.